Xinhua Chen

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Research Interests: Efficient AI systems through hardware–software co-design, with a focus on AI computing architectures, AI memory systems, and model–architecture co-evolution.

Education

Tsinghua University, Institute for Interdisciplinary Information Sciences (IIIS) *Sep. 2026 – Present*
Ph.D. in Computer Systems Architecture

- **Academic Service:** Reviewer for the International Conference on Learning Representations (ICLR).

Fudan University, College of Integrated Circuits & Micro-Nano Electronics *Sep. 2021 – Jun. 2026*
Bachelor of Engineering in Microelectronics

GPA: 3.78/4.00; ranked 13th out of 137 students

- **Honors:** Outstanding Graduate of Shanghai, 2026 (7 of 137 students); Xu Zengshou Scholarship, 2022 (1 of approximately 130 students).
- **Awards & Distinctions:** National Second Prize, National Mathematical Contest in Modeling ("HEP Cup"), 2022; Second-class Undergraduate Scholarship, 2023.
- **Research Experience:** Conducted research on fully homomorphic encryption accelerator scheduling, LLM-based RTL generation, and zero-shot HDR imaging; co-first author of ISEDA 2025 and GLSVLSI 2025, and co-author of CVPR 2024 Highlight.

Internship

Polar Bear Technologies *Jul. 2026 – Present*
Chip Architect *Advisor: Dr. Zhanhong Tan*

- **Responsibilities:** Investigate architectural options for the company's next-generation 3D-stacked near-memory processor and formulate research problems from hardware design challenges.
- **Achievements:** Developed this research into a manuscript submitted to **ASPLOS 2026** (under review).
- **Current Work:** Investigate HBF use cases and memory management strategies for LLM inference.

Compiler Team Member *Mar. 2026 – Jun. 2026*

- **Responsibilities:** Developed compiler optimizations for the company's in-house 3D-stacked near-memory processor, with responsibility for automatic weight-layout selection and GEMM scheduling for LLM inference.
- **Challenges Addressed:** Addressed DRAM row-buffer stalls by jointly modeling memory-access timing and compute pipelines to guide weight tiling, alignment, and scheduling.
- **Achievements:** Developed a lightweight layout search engine, **integrated into the company's compiler**, and formalized the work in an undergraduate thesis. Model-based evaluation showed **3.60–5.86× speedup** on large Linear operators over a row-unaware, row-major baseline, and **2.01× GEMM speedup** across GLM-4.7 MoE with only 0.63% DRAM storage overhead.

Duke University, Department of Electrical & Computer Engineering *Jun. 2025 – Sep. 2025*
Research Intern *Advisor: Prof. Yiran Chen*

- **Responsibilities: Led the diffusion language model research group and DPad project**, proposing suffix dropout for efficient inference and independently implementing the open-source codebase.
- **Achievements:** Co-first-author ICLR 2026 paper (DPad), based on research completed during the internship.

Shanghai Qi Zhi Institute *Mar. 2024 – Jun. 2025*
Research Intern *Advisor: Prof. Mingyu Gao*

- **Responsibilities:** Took over CROPHE and progressively assumed **project leadership**, developing memory-aware scheduling and unified processing elements for fully homomorphic encryption (FHE) accelerators.
- **Achievements:** Co-first-author paper at HPCA 2026 (CROPHE); co-author paper at DATE 2025 on a unified FHE vector processing unit.

Technical and Language Skills

Programming: C/C++, Python; **HDLs:** Verilog, Chisel; **Assembly:** RISC-V
Tools: Vivado, Quartus, gem5, Git, MATLAB, LaTeX
IELTS: 7.5 overall (Listening 7.5, Reading 9.0, Writing 6.5, Speaking 6.5)
GRE: 330 (Verbal 160, 84th percentile; Quantitative 170, 94th percentile)

Publications and Research Contributions

(† indicates co-first author and equal contribution)

[1] **Xinhua Chen**†, Sitao Huang†, Cong Guo†, Chiyue Wei, Yintao He, Jianyi Zhang, Hai "Hellen" Li, Yiran Chen. *DPad: Efficient Diffusion Language Models with Suffix Dropout.* International Conference on Learning Representations (**ICLR 2026**) [Link]

- **Overview:** A training-free inference method for diffusion language models that reduces redundant suffix attention through a sliding window and distance-based suffix dropout.
- **Contributions (Project Lead):** Led the project, proposed the suffix dropout technique to reduce interference from low-entropy suffix tokens, and independently developed the open-source codebase.
- **Results:** Up to 61.4× speedup over vanilla diffusion language models at comparable accuracy on LLaDA and Dream, with compatible decoding optimizations.
- **Highlights:** Received collaboration invitations from Ant Group, AMD, and Meta; covered by Synced (Jiqizhixin) and other technology media. The suffix dropout technique also informed improvements in diffusion language model training.

[2] **Xinhua Chen**†, Jiangbin Dong†, Hongren Zheng, Tian Tang, Mingyu Gao*. *CROPHE: Cross-Operator Dataflow Optimization for Fully Homomorphic Encryption Accelerators.* IEEE International Symposium on High-Performance Computer Architecture (**HPCA 2026**) [Link]

- **Overview:** An FHE accelerator combining unified processing elements with cross-operator scheduling to address memory bottlenecks through resource sharing and spatial/temporal pipelining.
- **Contributions:** Took over the project and progressively assumed leadership. Co-developed a memory-aware scheduler with loop reordering and tiling to reduce DRAM traffic and SRAM requirements; designed processing elements supporting forward/inverse NTT and MAC operations.
- **Results:** Reported 1.77×–4.86× speedup over the state-of-the-art accelerator designs evaluated in the paper.

[3] Jiangbin Dong, **Xinhua Chen**, Mingyu Gao*. *A Unified Vector Processing Unit for Fully Homomorphic Encryption.* Design, Automation and Test in Europe Conference (**DATE 2025**). [Link]

- **Overview:** A unified FHE vector processing unit that supports NTT and automorphism operations through shared hardware and an efficient inter-lane network.
- **Contributions:** Contributed to decomposing automorphisms into shifts and designing the inter-lane permutation network; co-proposed the unified vector processing unit.
- **Results:** Up to 9.4× area and 6.0× power savings for the inter-lane network; up to 1.2× area and 1.1× power savings for the full vector unit, relative to the designs evaluated in the paper.

[4] Ruoxi Zhu, Shusong Xu, Peiye Liu, Sicheng Li, Yanheng Lu, Dimin Niu, Zihao Liu, Zihao Meng, Li Zhiyong, **Xinhua Chen**, Yibo Fan*. *Zero-Shot Structure-Preserving Diffusion Model for High Dynamic Range Tone Mapping.* IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR 2024, Highlight**). [Link]

- **Overview:** A zero-shot HDR tone-mapping method that transfers a pretrained diffusion model to HDR inputs while preserving image structure, without paired HDR/LDR training data.
- **Contributions:** Evaluated tone-mapping quality using metrics including TMQI.

[5] Zihao Chen†, Zihan Lin†, **Xinhua Chen†**, Zhiyi Liu, Changxu Liu, Yuxuan Qiao, Yifei Feng, Junjie Zuo, Yifan Song and Fan Yang*. *Spec2Doc2RTL: RTL Generation from Specification with Natural Language Representation.* International Symposium of Electronics Design Automation (**ISEDA 2025**). [Link]

- **Overview:** A two-stage framework that translates specifications into RTL through natural-language design documents, using hierarchical decomposition and iterative feedback.
- **Contributions:** Contributed to recursive task decomposition and co-developed the open-source GPT-4 Turbo-based framework with iterative feedback for design-document and RTL generation.
- **Results:** Achieved **78.8% zero-shot pass@1** on Revisiting VerilogEval versus 60.3% for GPT-4 Turbo (**+18.5 percentage points**); demonstrated RISC-V CPU and NTT generation.

[6] Yi Sun†, **Xinhua Chen**†, Xinglong Yu, Wenxuan Zhang, Yifan Zhao, Honglin Kuang, Jun Han*. *Sliding-Window Scheduling to Exploit Hybrid-Bonding-Based Accelerators for Fully Homomorphic Encryption.* Great Lakes Symposium on VLSI (**GLSVLSI 2025**). [Link]

- **Overview:** Sliding-window scheduling for hybrid-bonding-based FHE accelerators that reduces data transfers and synchronization overhead through finer-grained dataflow optimization.
- **Contributions: Built the experimental framework and conducted nearly all experiments**.
- **Results:** Simulation showed **77.5% / 70.1% higher private-inference throughput** on ResNet-20 / ResNet-56 than H3 under identical hardware conditions.